# Unsupervised Learning II: Dimensionality Reduction

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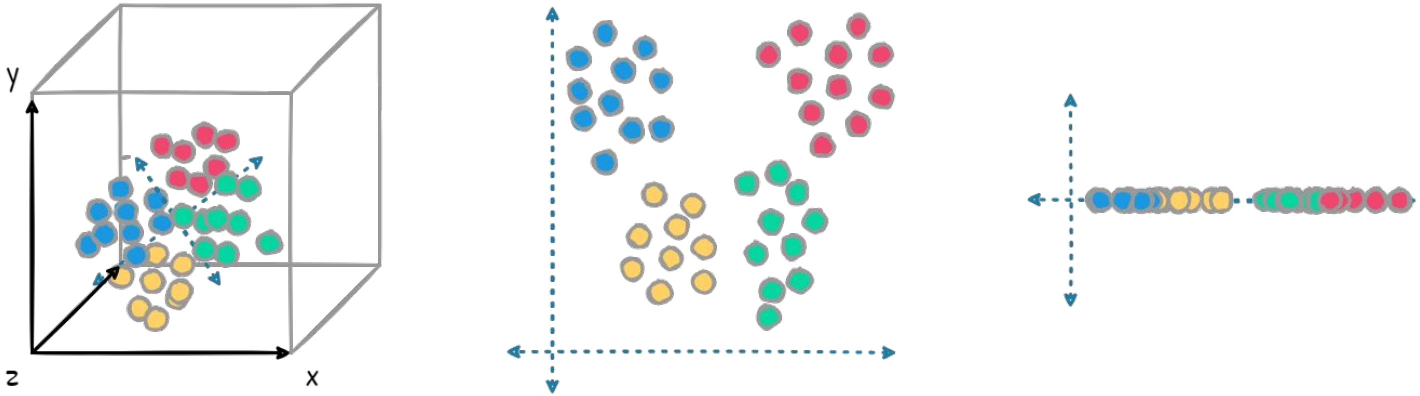
Date: 16th Feb 2026

# What is Dimensionality Reduction?

- **Reducing** the **number of features** while retaining **essential data structure**;

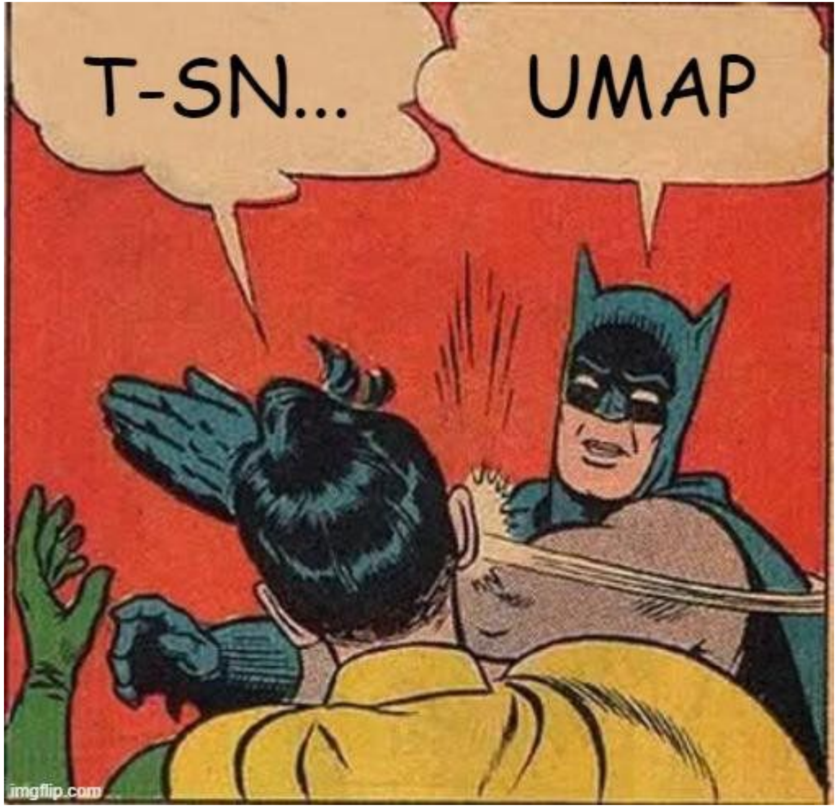



- **Importance:**
  - Curse of dimensionality: More dimensions can lead to sparse data and poor model performance;
  - Improved visualization and computation efficiency.

https://blog.roboflow.com/what-is-dimensionality-reduction/

# Today’s Objectives

- Understand the challenges posed by high-dimensional data;
- Key techniques:
  - PCA,
  - t-SNE,
  - UMAP.
- Use Cases;.
- Challenges:
  - Interpretability;
  - Determining dimensionality.

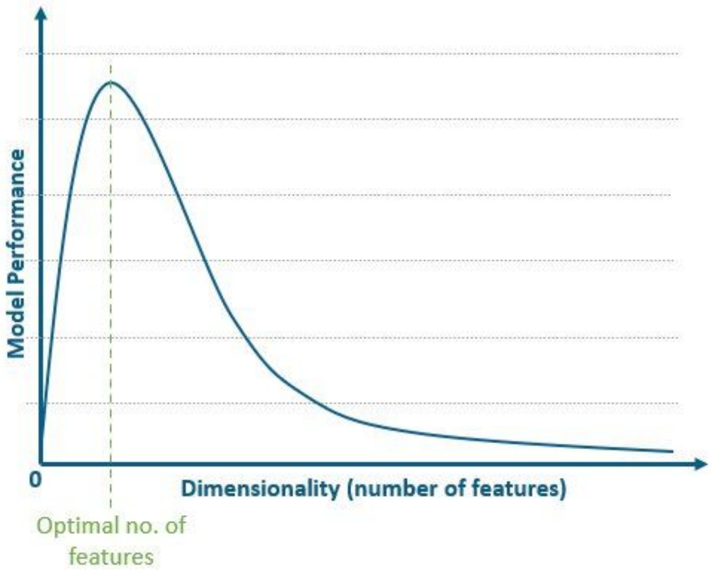



https://www.learndatasci.com/tutorials/applied-dimensionality-reduction-techniques-using-python/

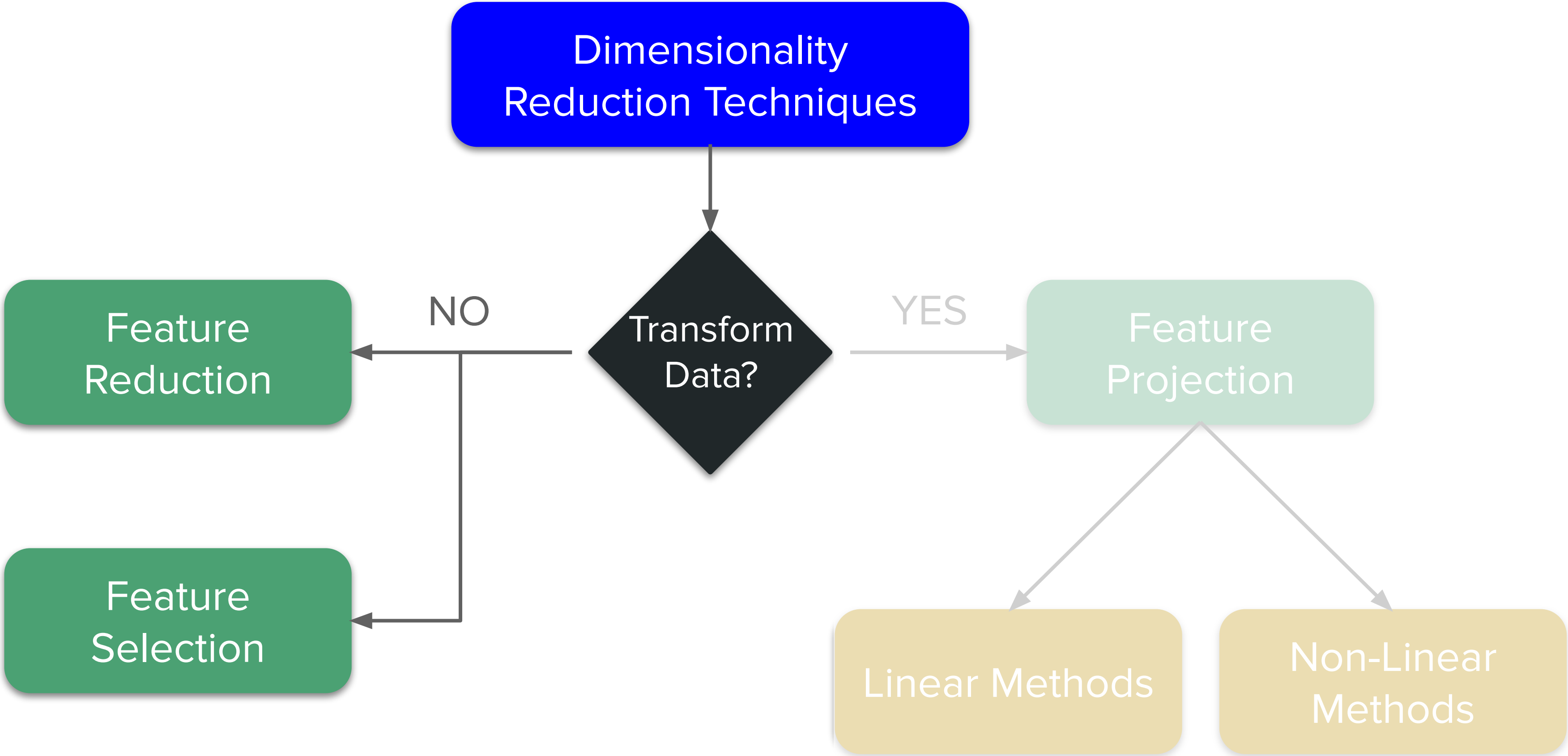
# Intro to Dimensionality Reduction

# The Curse of Dimensionality

- Simplifies data for visualization and analysis;
- Improves model performance by removing noise and redundancy;
- Enables feature selection or extraction for better interpretability.
- In high-dimensional spaces:
  - Distance measures become less meaningful.
  - Data points may all seem equidistant.
  - Models may overfit due to noise.




https://www.shiksha.com/online-courses/articles/curse-of-dimensionality/

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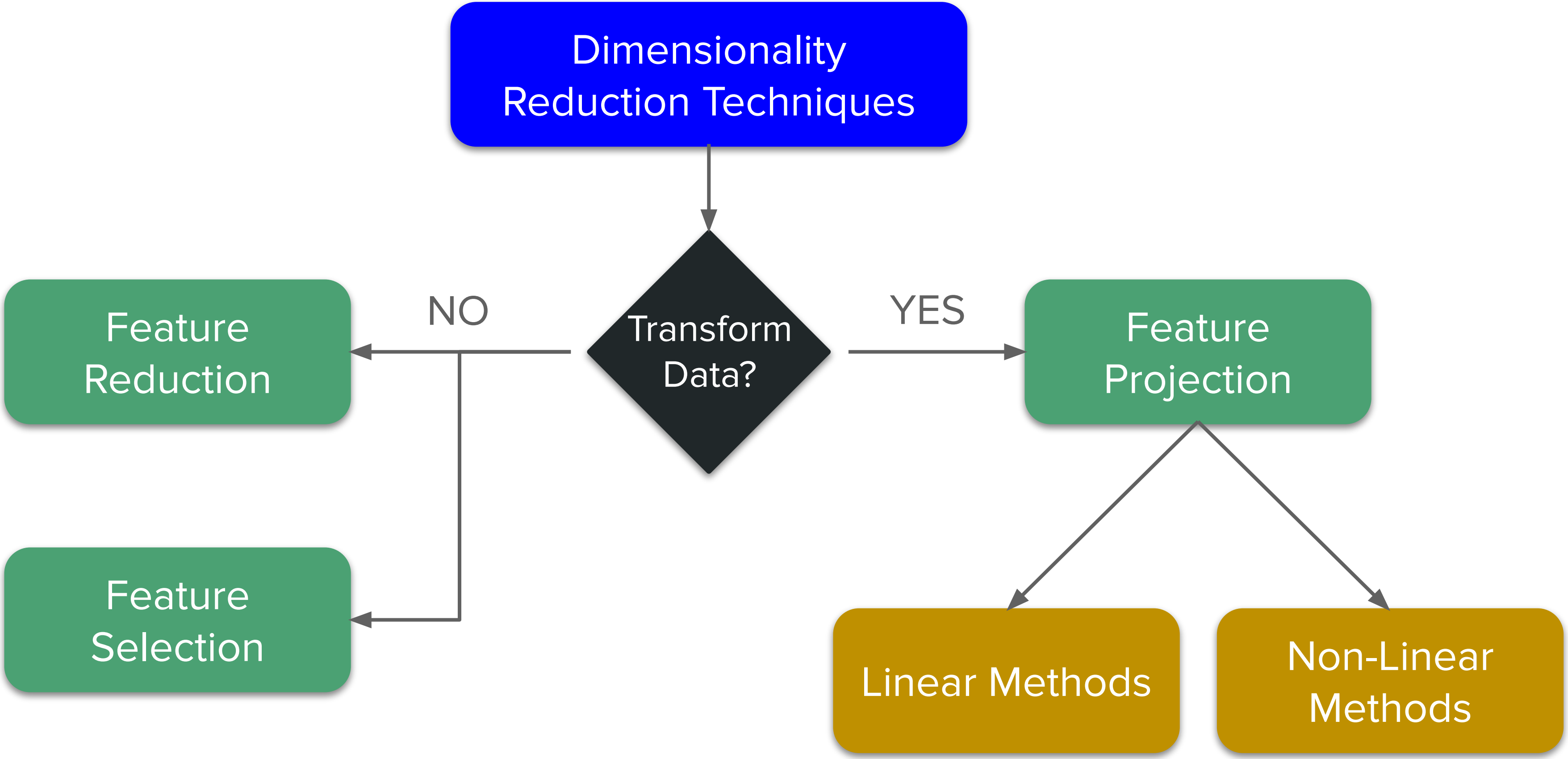


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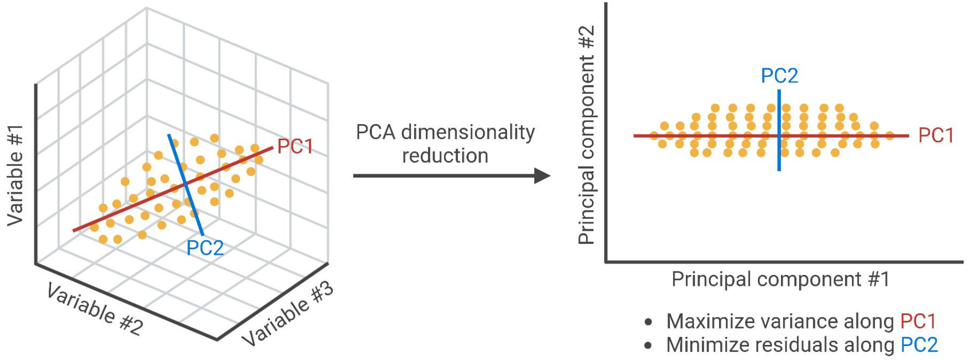
# Key Techniques

# Principal Component Analysis

- **Overview:**
  - Finds orthogonal axes (principal components) maximizing variance.
  - Projects data onto a lower-dimensional space.
- **Strengths: Linear**, fast, interpretable.
- **Weaknesses:** Assumes linear relationships, may miss non-linear patterns.

**Original data (high dimension)**




**Lower-dimensional embedding**

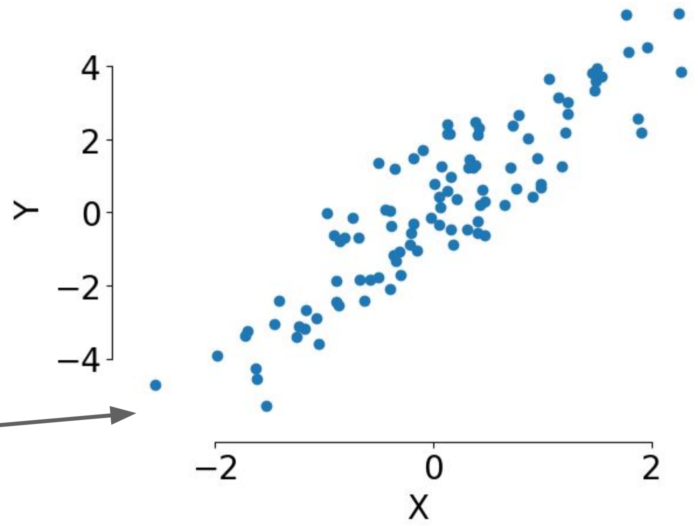
https://www.biorender.com/template/principal-component-analysis-pca-transformation

# Principal Component Analysis

**Algorithm:**

1. **Compute the Covariance Matrix**:
   - Calculate the covariance matrix to measure how variables in the dataset vary together; example:

```
[[1.02608749 2.16986562]
 [2.16986562 5.65646178]]
```



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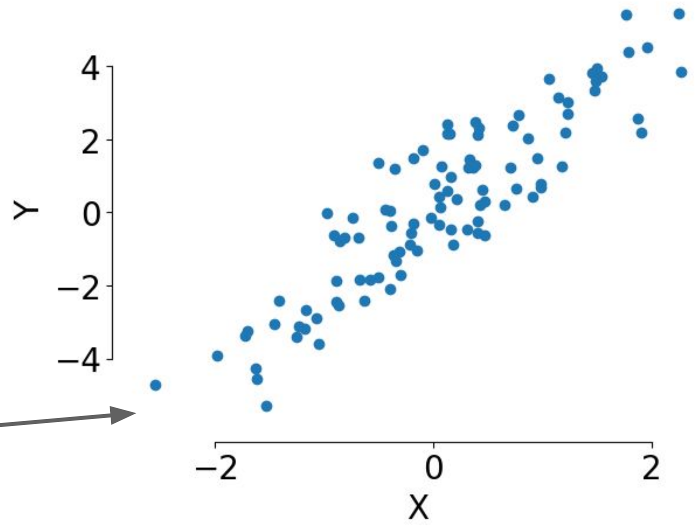


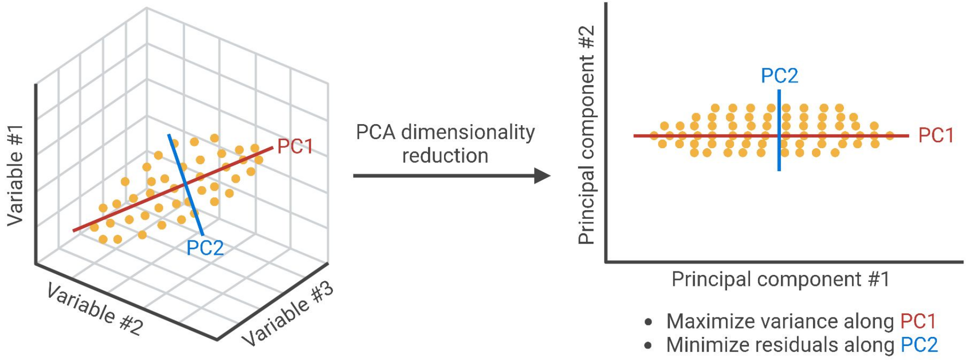

2. **Compute Eigenvalues and Eigenvectors**:
   - Calculate the eigenvalues and eigenvectors of the covariance matrix. These represent the directions (principal components) in which the data varies the most.
3. **Sort Eigenvalues**:
   - Sort the eigenvalues in descending order. The larger the eigenvalue, the more variance is captured in that direction.

# Principal Component Analysis

## Algorithm:

4. **Select Top Principal Components**:
    - Choose the top N eigenvectors (principal components) based on the largest eigenvalues. These form the new axes for the data.

**Original data (high dimension)**

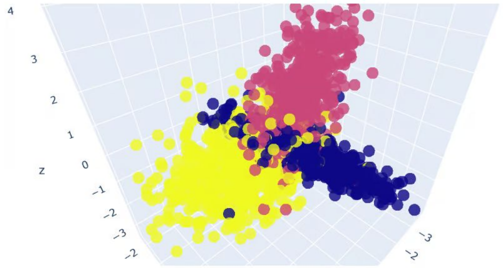



**Lower-dimensional embedding**

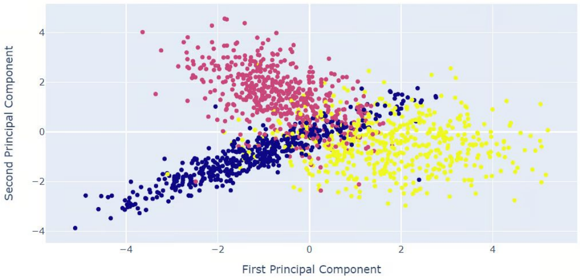
https://www.biorender.com/template/principal-component-analysis-pca-transformation

# t-Distributed Stochastic Neighbour Embedding (t-SNE)

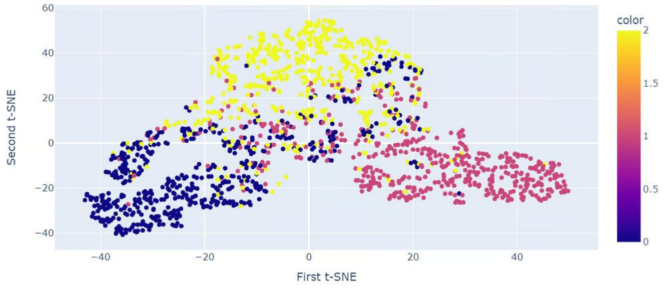
- **Overview:**
  - Preserves local structure of data in a lower-dimensional space.
  - Great for visualization (e.g., clusters).
- **Strengths:** Captures **non-linear** relationships, good for exploratory analysis.
- **Weaknesses:** Computationally expensive, parameters sensitive.



**Original Data (3D)**




**PCA**




**t-SNE**

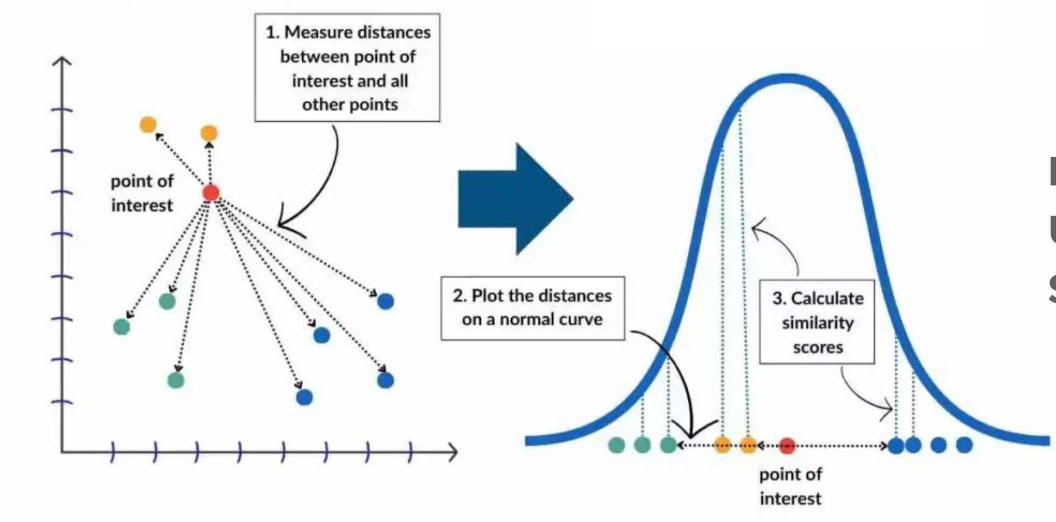
https://www.datacamp.com/tutorial/introduction-t-sne

# t-Distributed Stochastic Neighbour Embedding (t-SNE)

## Algorithm:

1. **Measure Similarity**:
   - Evaluates the similarity between pairs of data points in the high-dimensional space.

**Original Data Distribution**




**Normal Distribution Used to Calculate Similarity**

https://spotintelligence.com/2023/12/22/practical-guide-t-sne/

# t-Distributed Stochastic Neighbour Embedding (t-SNE)

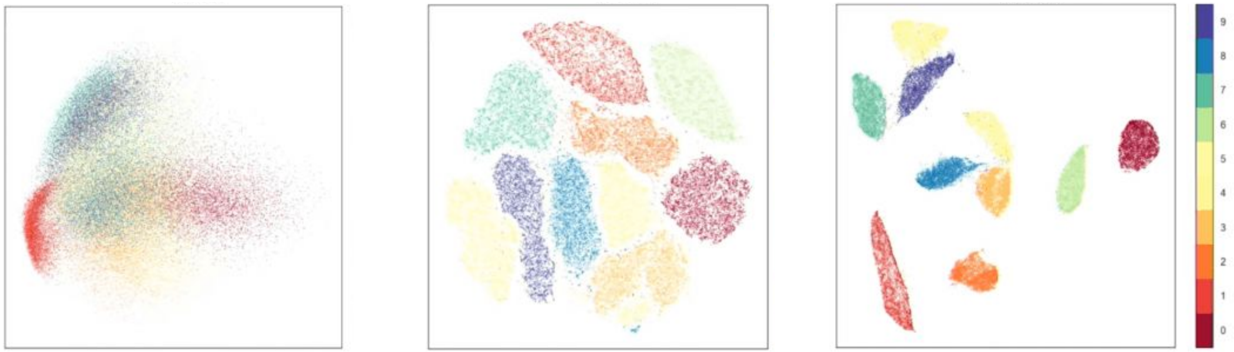
**Algorithm:**

1. **Measure Similarity**:
   - Evaluates the similarity between pairs of data points in the high-dimensional space.
2. **Map to Lower Dimensions**:
   - Represents these similarities in a simplified space while preserving the relationships between points.
3. **Iterative Adjustment**:
   - Adjusts point positions in the lower-dimensional space to closely match the original similarities
   - Achieved by minimizing the divergence between the original high-dimensional and lower-dimensional probability distribution.

# Uniform Manifold Approximation and Projection (UMAP)

- **Overview:**
  - Similar to t-SNE but more efficient and better for larger datasets.
  - Preserves both local and global data structures.
- **Strengths:** Faster, scales well, captures non-linear relationships.
- **Weaknesses:** Interpretability and parameter sensitivity.

**MNIST Digits**



**PCA** **t-SNE** **UMAP**

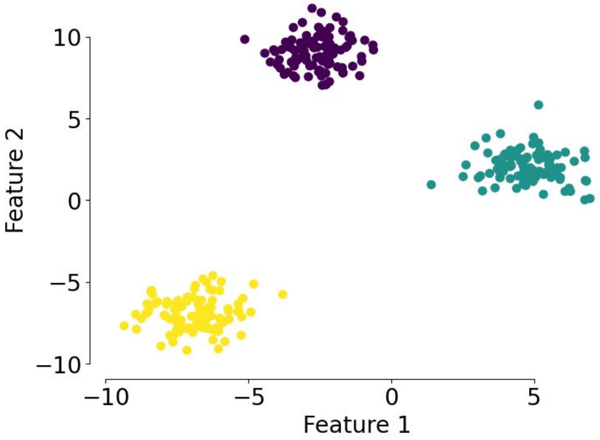
https://meta.caspershire.net/umap/

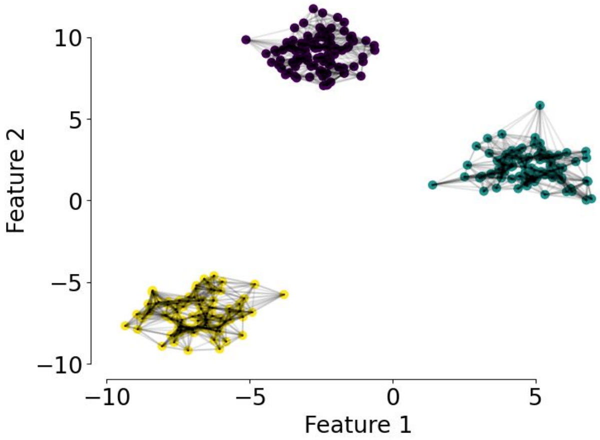
# Uniform Manifold Approximation and Projection (UMAP)

**Algorithm:**

1. **Construct a Nearest-Neighbor Graph:**
   - Build a graph of data points where each point is connected to its k-nearest neighbors using a distance metric (e.g., Euclidean distance).

**Original Data**

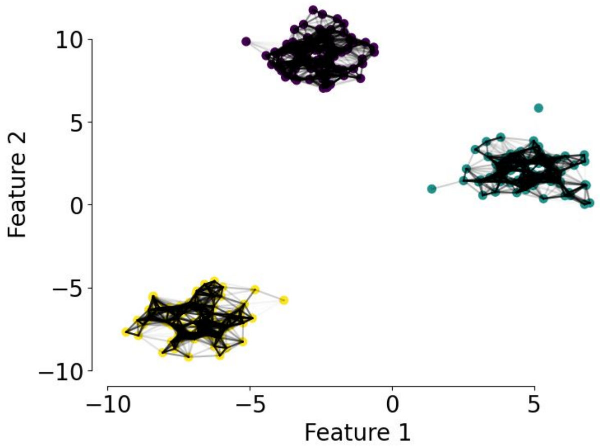


# Uniform Manifold Approximation and Projection (UMAP)

**Algorithm:**

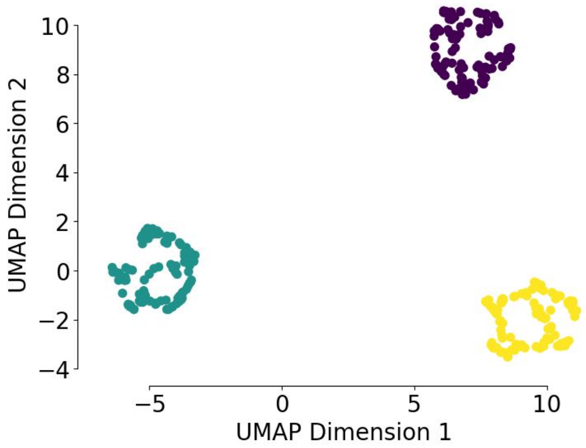
1. **Construct a Nearest-Neighbor Graph:**
    - Build a graph of data points where each point is connected to its k-nearest neighbors using a distance metric (e.g., Euclidean distance).
2. **Define Local Neighborhoods:**
    - Assign a weight to each edge in the graph based on the similarity between the connected points. This is done using a distance-based kernel function (often Gaussian or other smooth functions).



# Uniform Manifold Approximation and Projection (UMAP)

**Algorithm:**

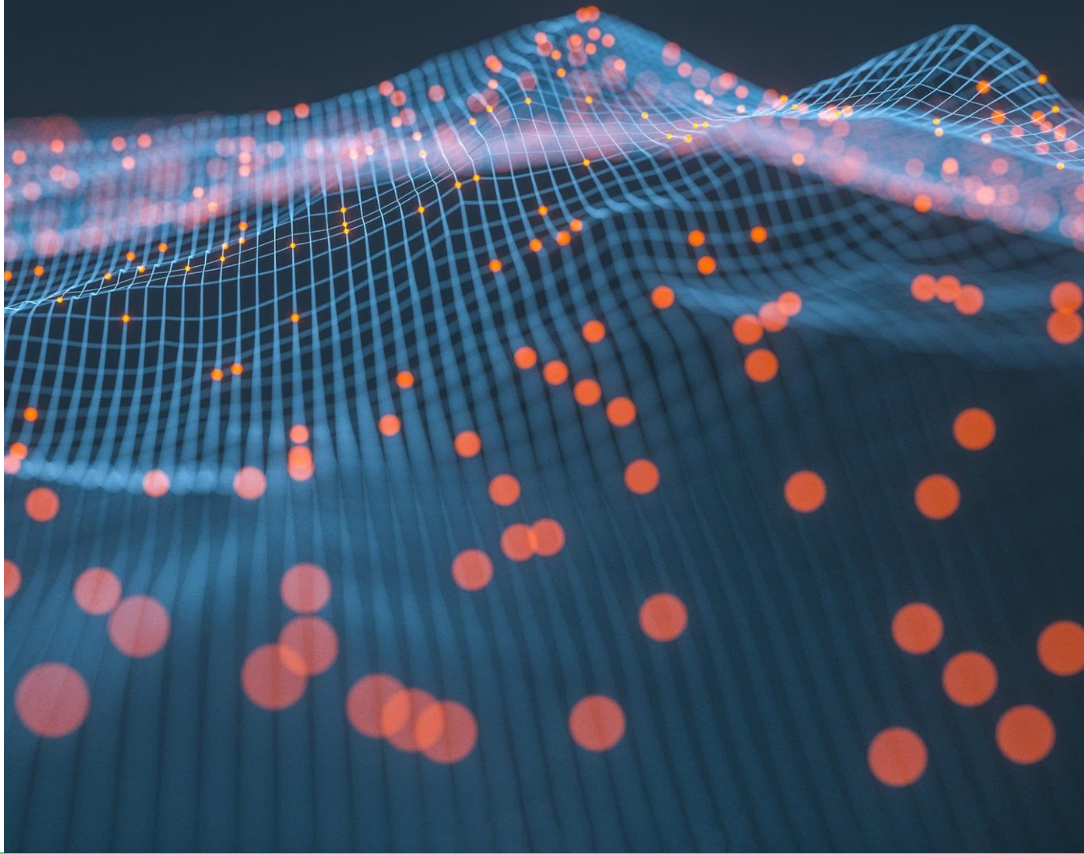
3. **Construct a Manifold Representation:**
    - Use the graph to model the local manifold of the data.
4. **Embed into Low-Dimensional Space:**
    - Project the high-dimensional data onto a low-dimensional space.
5. **Optimize Embedding:**
    - Ensure that both local neighborhood structures and larger global relationships are preserved.



# Use Cases

# Dimensionality Reduction Use Cases

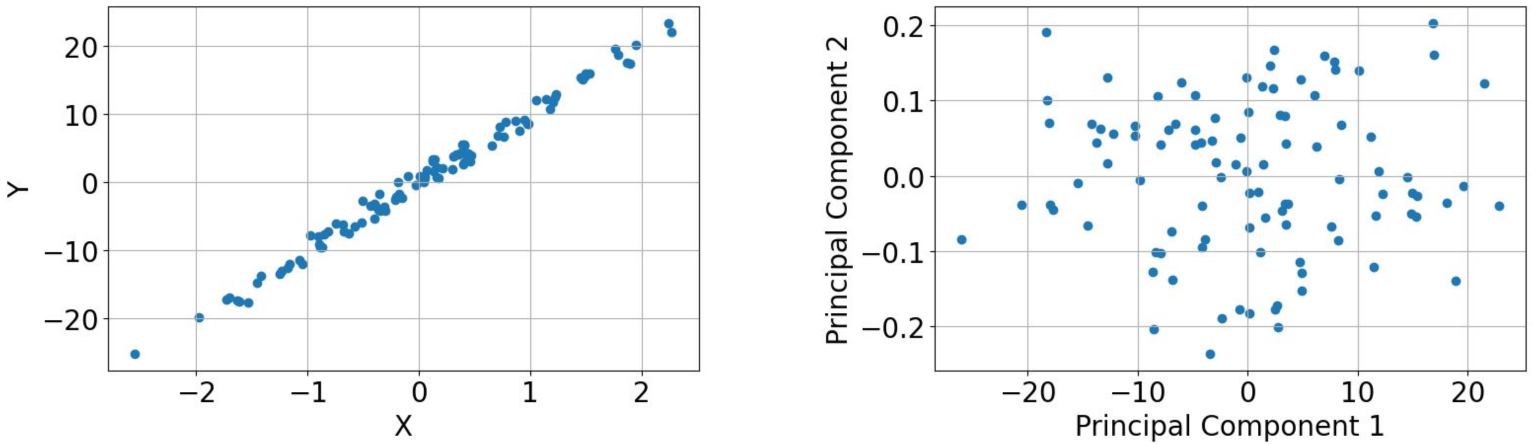
- **Clinical Bioinformatics:**
    - Gene expression,
    - Radiomics,
    - Multi-omics data integration,
    - Patient stratification,
    - Survival analysis.
- **Other Domains:**
    - Customer segmentation,
    - Speech/audio processing,
    - Social media analysis,
    - Financial market analysis,
    - Recommendation systems.



# Challenges

# Interpretability

- Reduced dimensions can obscure meaning behind features.
  - Example: PCA transforms features into principal components, losing original context.




- Carefully link reduced features to original variables where possible!

# Determining Dimensionality

- **Balancing trade-offs:**
    - Too few dimensions ➔ information loss.
    - Too many dimensions ➔ overfitting or inefficiency.

# Determining Dimensionality

- **Balancing trade-offs:**
  - Too few dimensions ➜ information loss.
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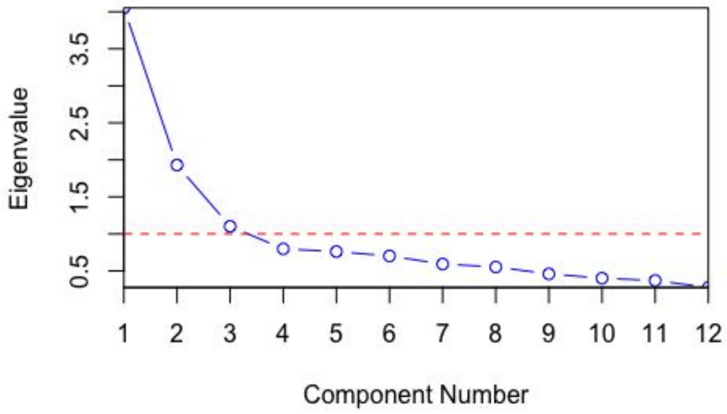
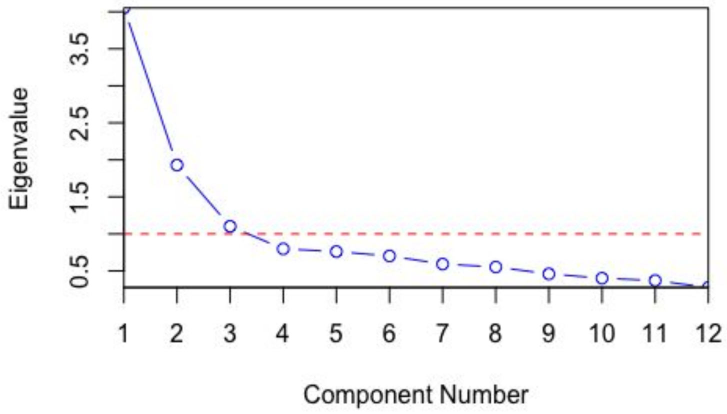
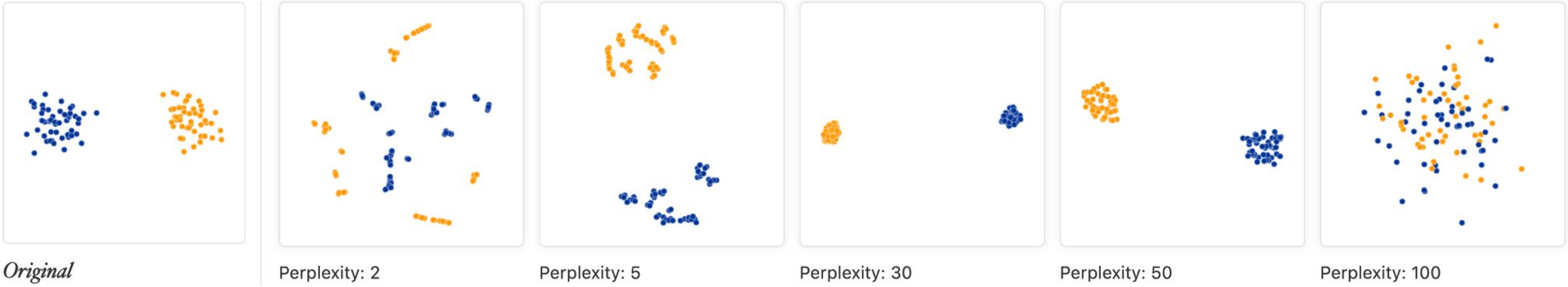
- **Tools to decide:**
  - Scree plots for PCA,



# Determining Dimensionality

- **Balancing trade-offs:**
  - Too few dimensions ➔ information loss.
  - Too many dimensions ➔ overfitting or inefficiency.
- **Tools to decide:**
  - Scree plots for PCA,
  - Perplexity tuning for t-SNE,
  - Neighbour tuning for UMAP.




https://distill.pub/2016/misread-tsne/

# Summary and Q&A

# Summary - Key Techniques

| Characteristic | PCA | t-SNE | UMAP |
|---|---|---|---|
| **Type** | Linear dimensionality reduction | Non-linear dimensionality reduction | Non-linear dimensionality reduction |
| **Goal** | Maximize variance | Preserve local pairwise similarities | Preserve both local and global structures |
| **Best Used For** | Data with linear structure, noise reduction | Complex, high-dimensional data (e.g., NLP, clustering) | Complex, high-dimensional data with both local and global structures |
| **Output** | Principal components | Low-dimensional representation (2D/3D) | Low-dimensional representation (2D/3D) |

# Summary - Key Techniques

| Characteristic | PCA | t-SNE | UMAP |
|---|---|---|---|
| **Use Cases** | Noise reduction, feature extraction | Clustering, anomaly detection, visualization | Visualization, clustering, outlier detection |
| **Computational Intensity** | Low | High | Moderate |
| **Interpretation** | Easy to interpret | Harder to interpret | Easier to interpret than t-SNE |
| **Preserved Information** | Variance | Local pairwise similarities | Local and global structure |
| **Scalability** | High (works well for large datasets) | Low (slow for large datasets) | High (faster and more scalable than t-SNE) |

Questions?


WHAT IF I TOLD YOU
PCA ALLOWS PROJECTING HIGH
DIMENSIONAL DATA INTO LOWER DIMENSIONS